



Overview

Team

We are the team Mathimazierer from Lessing-Gymnasium Neu-Ulm in Germany. Our team consists of four members, and each of us has a specific role: Theo is responsible for programming, Dominik works on 3D design, Emanuel designs the PCBs, and Mika takes care of the hardware and soldering.

School

At our school, students can join the robotics club starting in 5th grade. In the beginning, they learn the basics using simple LEGO robots that only need to drive in a straight line. In 6th grade, students start competing in RoboCup for the first time, usually in leagues such as Rescue Line Entry or Soccer Entry Standard Kit. From 7th grade onward, students are free to choose which league they want to participate in. This year, our school has been especially successful: we won the German Open in the Soccer Open, OnStage, and Soccer Lightweight International leagues. Now, we are heading to the World Championship with three teams.

History

We have been participating in RoboCup since 2022, and every year since then we have taken part in the German Open and South German Open. We also took part in the European Open twice, where we achieved 2nd and 3rd place.



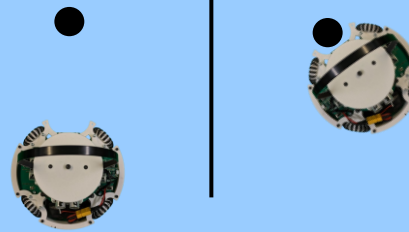
School



Team

Tactics

Goal



To shoot as many goals as possible, we utilize a 360° camera and a light barrier. These allow us to accurately detect both the ball and the goal.

Ball not in possession: When the ball is not in our ball-capturing zone, the robot aligns itself with the field to optimize its defensive positioning.

Ball in possession: As soon as the robot captures the ball, it drives directly towards the goal, rather than alongside it, to ensure a straight and efficient shot.

Hardware

IR-Board

To detect the ball, we use 16 infrared sensors arranged in a circle with 22.5° spacing between each sensor. These sensors are mounted on a dedicated IR board and connected to the mainboard via cables.

Bottom-Board

For line detection, we built a special bottom board that includes 32 LEDs, each paired with 2 phototransistors, totaling 64 phototransistors. These are grouped into sets of 16 and connected to 4 multiplexers. The multiplexed signals are routed via cables directly to our mainboard. The bottom board is round and includes a cross-shaped pattern to provide maximum information about the field lines.

Mainboard

This year, we are using four brushless motor drivers from Maxon to control our four motors. The motor drivers are powered directly from our 11.1V battery via a power switch. Remaining power is converted into 3.3V and 5V through voltage regulators for the rest of the system.

Our mainboard also features a Teensy 4.1 microcontroller, which handles all signal processing and control, as well as a BNO055 orientation sensor for detecting the robot's heading on the field.

Other Components

For the first time, we are using a 360° camera system. This consists of an OpenMV H7 camera that points into a curved mirror, allowing a full view of the surroundings. We also added ultrasonic sensors to assist with orientation and distance measurement on the field.

Programs we used



Improvement

